KING SALMAN INTERNATIONAL UNIVERSITY

Field of Engineering and Computer Engineering

All Program

Course : Mathematics 1 (MAT 111)

Lecture 1: Introduction

Dr : Ahmed Ali

Date : / 9 /2026

Course Administration

Instructor: Dr. Ahmed Ali Mohamed
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Office Hours:
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Textbook:

- Robert T. Smith and Roland B. Minton "**Calculus(Early Transcendental Function)**" 5th. Edition, Prentice Hall, 2010.

Additional Readings:

- Howard Anton, Irl Bivence Stephan Davis"**Calculus(Early Transcendental)**" 10th. Edition,, inc., New Yourk
- Erwin Kreyszig "**Advanced Engineering Mathimatics**" Saunders Collage Publishing.

Lecture Hall Rules

- Students are not allowed to enter the lecture hall after the beginning of the lecture
- Students should maintain discipline, perfect order, and strict silence inside the lecture hall
- Smoking, eating, drinking, chewing gum, ***personal*** use of mobile-phones, camera, iPods, MP3 players or any other electronic gadgets in the lecture hall is strictly prohibited
- No student is allowed to leave the lecture room without the permission of the lecturer or until the class is dismissed

Course Description

The course consists of two Main Parts:

- The first part includes Functions with their graph, Limits and continuity, Differentiation of functions, and application of derivatives.
- The second part includes some concepts of linear algebra; analyzing the function with partial fraction, matrices and their properties, inverse of the matrix, and solution of a system of simultaneous equations with matrix inversion.

Course prerequisites:

Co-requisites for this course :

Course Objectives

- Introduce the fundamental of functions, graphing, limits and continuity.
- Study the differentiation of the functions.
- Introduce some applications of the differentiation.
- Analyze the function with partial fraction.
- Study some concepts of linear algebra to find the inverse of a matrix.
- Solve a system of simultaneous equations using matrix inversion.

Course contents

Calculus part

- Functions
- Limits
- Continuity
- Derivatives
- Applications of differentiation

Linear Algebra part

- Systems of Linear Equations
- Theory of Matrices
- Applications in Systems of Equations

textbook: R. Smith, R. Minton, z. Rafhi, Calculus: Early Transcendental Functions, 5th Edition, McGraw-Hill Education.

Grade Distribution

No.	Assessment method	Points	Weeks
1	Quizzes	15	4-6-11-13
2	Mid Term exam	20	8
3	Assessments	15	6-13
5	Final Exam	50	15

Course Policies:

1. Homework Submission:

   Homework solutions should be submitted on the due date.

2. Late homework policy:

   On time, in the class (100%), **same day after class (90%), next day (70%), 2 days (50%), 3 days or more (0%).**

3. Collaboration:

   You are encouraged to discuss the assigned problems/projects with your classmates. But you are **not allowed to talk about the final solution itself or to show your solution to others.** Every student has to prepare his/her solution independently.

4. Preparing the final solution:

   Please write your solution in a clear, readable, and concise form. Every **answer should be fully justified**.



- definition : R = set of all rational and irrational numbers (real numbers)

  e.g. $\pi \in R$, $\pi \notin Q$, $\sqrt{2} \in R$, $\sqrt{2} \notin Q$,$\sqrt{64} \in R$, $\sqrt{64} \in N$

- definition : subsets of sets : $A \subseteq B$ if and only if every $a \in A$ obeys $a \in B$

  note : $N \subseteq Z \subseteq Q \subseteq R$

- Logical symbols: $\wedge$ (AND), $\vee$ (OR)
- S1 $\wedge$ S2: both statements S1 and S2 are true
- S1 $\vee$ S2: either S1 is true, or S2 is true, or both are true
- Logical consequences: S1 $\Rightarrow$ S2 means ‘if statement S1 is true then also statement S2 is true’

Numbers

- Natural numbers: $\mathbb{N} = \{1, 2, 3, 4, \dots\}$
- Whole numbers: $\mathbb{N}^* = \{0, 1, 2, 3, 4, \dots\}$
- Integer numbers: $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
- Rational numbers: $\mathbb{Q} = \{\boldsymbol{n/m};\ n, m \in \mathbb{Z};\ m \neq 0\}$ Examples: $\frac{1}{2} = 0.5,\ \frac{4}{3} = 1.\overline{3},\ \frac{-10}{5} = -2$
- Irrational numbers: $\mathbb{Q}' = \{e, \pi, \sqrt{2}, \dots\}$
- Real numbers: $\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}' =]-\infty, +\infty[$

$\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{Q}$ contain discrete data values, while $\mathbb{R}$ contains continuous data values.

The interval

- Closed interval: line segment that includes both end points

  $[a, b] = \{x \in R \mid a \leq x \leq b\}$

- Open interval: line segment that does not include end points

  $(a, b) = \{x \in R \mid a < x < b\}$

- Semi-open (or semi-closed) interval, exactly one endpoint is included

  $[a, b) = \{x \in R \mid a \leq x < b\}$ or $(a, b] = \{x \in R \mid a < x \leq b\}$

Powers of real numbers

In an expression of the form x^n we call x the base and n the power.

First define natural powers of real numbers. Let n ∈ N and x ∈ R, $x \neq 0$:

definition $x^0 = 1$ definition $x^n = x.x.\ldots x$ (n−fold product, for n > 0)

Generalize to integer powers, by giving a definition for negative powers. Let n ∈ N, n > 0:

definition $x^{-n} = \frac{1}{x.x.\ldots.x}$ (n−fold product in denominator

Three basic laws of manipulation, let x ∈ R and n,m ∈ Z:

- first law : $x^n . x^m = x^{n+m}$
- second law : $\frac{x^n}{x^m} = x^{n-m}$
- third law :$(x^n)^m = x^{n.m}$

Generalize to fractional powers of positive real numbers x ∈ R_+, by giving a definition for

powers of the form $1/n$ with n ∈ N, n > 0:

definition $x^{1/n} = \sqrt[n]{x}$ (n^{th} root)

Here $\sqrt[n]{x}$ is the number y ∈ R_+ with the property that $y^n = x$.

Verify that the above laws of manipulation still hold in the case of fractional powers.

For example, let m, n, ℓ ∈ N_+:

$$a^{l/n} = (\sqrt[n]{a})^l$$

$$a^{q+l/n} = a^q . (\sqrt[n]{a})^l$$

Equation

LHS = RHS

Left Hand Side

Right Hand Side



Inequality

LHS $<$ **RHS**

$\leq$

$>$

$\geq$



For the line $y = f(x) = 2x - 7$ **,**

1 Evaluate $f(4)$.

$$y = f(4) = 2(4) - 7 = 1$$

$x = 4$

$y = ?$

(one value)

2 Solve the equation $2x - 7 = 5$.

$y = 5$

$x = ?$

$$2x = 12 \quad \Rightarrow \quad x = 6$$

Solution Set: $x = \{6\}$

3

Solve the inequality $2x - 7 > 5$.

$y > 5$

x : ?

$$2x > 12 \quad \Rightarrow \quad x > 6$$

$$\therefore x \in]6, \infty[\equiv \textbf{Solution Set}$$

4

Solve the inequality $-3 \leq 2x - 7 \leq 1$.

$-3 \leq y \leq 1$

x : ?

$$4 \leq 2x \leq 8 \quad \Rightarrow \quad 2 \leq x \leq 4$$

$$\therefore x \in [2, 4] \equiv \textbf{Solution Set}$$

Note that

Taking the reciprocal or multiplying by (-1) of both sides will change the direction of the inequality.

1. $0 < \frac{1}{x} < 3 \quad \Rightarrow \quad \infty > x > \frac{1}{3} \quad \Rightarrow \quad x \in \left]\frac{1}{3}, \infty\right[$

2. $-x \geq 2 \quad \Rightarrow \quad x \leq -2 \quad \Rightarrow \quad x \in \left]-\infty, -2\right]$

Types of intervals

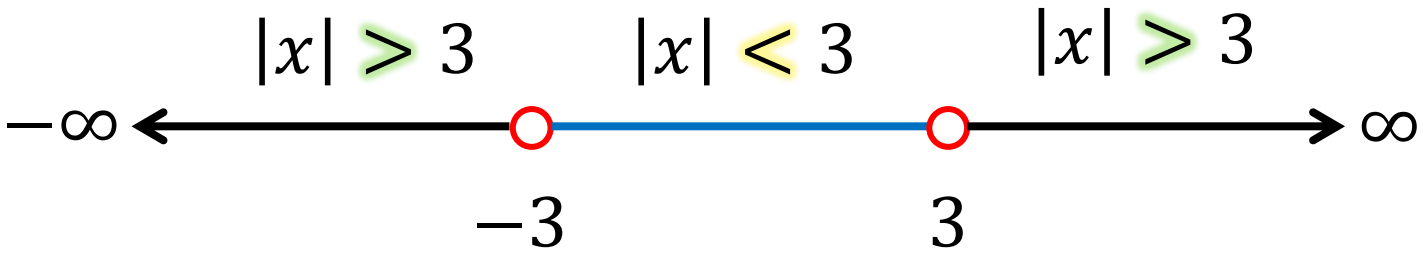
	Notation	Set description	Type	Picture
Finite:	(a, b) or $]a\,,b[$	$\{x \mid a < x < b\}$	Open	
	$[a, b]$	$\{x \mid a \leq x \leq b\}$	Closed	
	$[a, b)$ or $[a\,,b[$	$\{x \mid a \leq x < b\}$	Half-open	
	$(a, b]$ or $]a\,,b]$	$\{x \mid a < x \leq b\}$	Half-open	
Infinite:	(a, ∞) or $]a\,,\infty[$	$\{x \mid x > a\}$	Open	
	$[a, \infty)$ or $[a\,,\infty[$	$\{x \mid x \geq a\}$	Closed	
	$(-\infty, b)$ or $]-\infty, b[$	$\{x \mid x < b\}$	Open	
	$(-\infty, b]$ or $]-\infty, b]$	$\{x \mid x \leq b\}$	Closed	
	$(-\infty, \infty)$ or $]-\infty, \infty[$	$\mathbb{R}$ (set of all real numbers)	Both open and closed	

Note that

1. $|x| < 3 \Rightarrow -3 < x < 3 \Rightarrow x \in \,]-3\,,3\,[$

2. $|x| > 3 \Rightarrow x > 3 \quad \text{or} \quad x < -3$

$$x \in \,]-\infty\,,-3[\ \cup\ ]3\,,\infty[$$



Solve the equation $x^2 - 4x + 3 = 0$.

Factorization

$$(x - 1)(x - 3) = 0$$

Solution Set: $x = \{1 , 3\}$

OR

The general low

$$a x^2 + b x + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Solve the equation $\boldsymbol{x^2 - 4\,x + 3 = 0}$.

Formula

$$a = 1 \qquad b = -4 \qquad c = 3$$

$$x = \frac{-b \pm \sqrt{b^2 - 4\,a\,c}}{2\,a} = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(3)}}{2(1)} = \frac{4 \pm 2}{2}$$

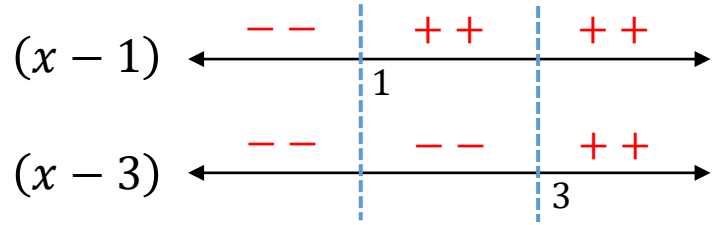
Solution Set ≡ Equation Roots ≡ Zeros of Function $\boldsymbol{f(x) = x^2 - 4\,x + 3}$:

$$x = \{1\,,\,3\}$$

Solve the inequality $x^2 - 4x + 3 < 0$.

Factorization

$$(x-1)(x-3) < 0$$



Solution Set: $x \in]1\,,\,3[$

OR

Completing the square

$$(x-2)^2 - 1 < 0$$

$$(x-2)^2 < 1 \xrightarrow{\sqrt{\ }} |x-2| < 1 \Rightarrow -1 < x-2 < 1$$

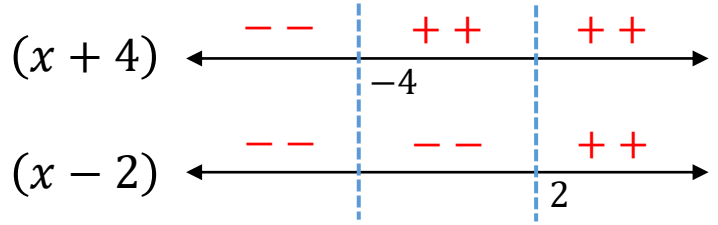
$$\therefore 1 < x < 3$$

Solution Set: $x \in]1\,,\,3[$

Solve the inequality $x^2 + 2x - 8 \geq 0$.

Factorization

$$(x-2)(x+4) \geq 0$$



Solution Set: $x \in]-\infty, -4] \cup [2, \infty[$

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THANK YOU

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